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RESEARCH ARTICLE

Big Data Analytics in Library Services With AI: Personalized Content Recommendations and Catalog Optimization

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ABSTRACT This paper presents an AI-driven framework designed to enhance user engagement and optimize catalog management in digital libraries. The framework integrates Variational Autoencoder (VAE)-based personalized recommendations with Adam optimizer and Lookahead mechanism for catalog optimization. The VAE model effectively learns latent representations of user-item interactions, providing personalized content recommendations. For catalog optimization, the Adam optimizer with Lookahead stabilizes convergence and refines inventory selection, leading to more efficient resource allocation and reduced costs. Experimental results from a large-scale dataset demonstrate that the proposed approach outperforms traditional methods, achieving significant improvements in recommendation accuracy and user engagement. It reduces the number of low-demand items while enhancing overall catalog efficiency. The proposed framework provides a scalable and adaptable solution for digital libraries, ensuring both user satisfaction and effective resource management. Future work will explore hybrid models incorporating Natural Language Processing (NLP) to improve content understanding and further enhance recommendation quality.

INDEX TERMS Variational autoencoder (VAE), catalog optimization, Adam optimizer, Lookahead mechanism, personalized recommendations, digital libraries.

I. INTRODUCTION

The rapid expansion of digital libraries has transformed the way users access, interact with, and consume knowledge. With the increasing digitization of books, research papers, and multimedia content, modern libraries have evolved into vast repositories of information. However, this exponential growth has introduced challenges in content discoverability, user engagement, and catalog management [1]. Users often struggle to find relevant materials due to information overload, while library administrators face difficulties in efficiently curating collections to match demand patterns. The inefficiencies in recommendation and catalog optimization lead to suboptimal user experiences and resource allocation [2]. Traditional recommendation systems, based on collaborative filtering (CF) and content-based approaches, have been widely used in digital libraries [3]. Collaborative filtering relies on user-item interaction data to predict preferences, while content-based methods use item attributes to make recommendations. Despite their success, these approaches suffer from several limitations, including:

1) Cold-start problem – New users and items have insufficient interaction data, making accurate recommendations difficult.
2) Sparsity – The interaction matrix between users and items is often sparse, leading to unreliable preference predictions.
3) Limited personalization – Traditional models struggle to capture complex, non-linear relationships in user-item interactions, leading to generic recommendations.

Another significant challenge in digital library management is catalog optimization. Libraries must maintain a balance between offering a diverse selection of materials and optimizing inventory to meet user demand while staying

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within budget constraints. Static inventory management approaches often result in inefficient resource allocation, where low-demand items occupy valuable storage, while high-demand items experience shortages [4]. Traditional optimization techniques lack adaptability and fail to incorporate real-time user engagement data, leading to suboptimal decision-making. To address these challenges, Artificial Intelligence (AI) and deep learning have emerged as powerful tools for transforming library services [5]. AI-driven models can leverage large-scale user interaction data to enhance content recommendations and optimize catalog management dynamically [6]. In this paper, we propose an AI-driven framework integrating deep learning-based recommendation systems with advanced optimization techniques to improve user engagement and resource allocation efficiency. The digital transformation of libraries has led to vast, information-rich repositories, but it also introduced major challenges in content discoverability, personalized recommendations, and catalog efficiency. Traditional methods like collaborative and content-based filtering suffer from cold-start issues, data sparsity, and limited personalization. Moreover, static inventory strategies fail to adapt to dynamic user demands, resulting in inefficient resource allocation. This evolving landscape necessitates AI-driven solutions to optimize both user engagement and catalog management in modern digital libraries.

The existing literature on digital library recommendation systems offers several advantages, primarily through the use of collaborative filtering (CF) and content-based approaches, which are well-established and widely adopted. These models are relatively simple to implement, computationally efficient, and effective in environments with sufficient interaction data, making them suitable for large-scale deployment in digital libraries. They enable quick prototyping and provide interpretable results based on either user behavior or item attributes. However, despite these strengths, they exhibit notable limitations. Chief among these are the cold-start problem—where new users or items lack sufficient data for accurate prediction—and the sparsity of user-item interaction matrices, which undermines recommendation quality. Additionally, traditional methods are constrained by their linear modeling assumptions and limited capacity to capture complex, non-linear patterns in user preferences, resulting in less personalized and more generic recommendations. Moreover, they often lack adaptability and fail to incorporate real-time behavioral data, which diminishes their effectiveness in dynamic environments such as digital libraries that constantly evolve in terms of content and user engagement. Deep learning models have shown remarkable success in recommendation systems by learning latent representations of users and items. Variational Autoencoders (VAE), a generative deep learning model, has been particularly effective in learning probabilistic latent factors from sparse interaction data [7]. Unlike traditional matrix factorization techniques, VAEs can capture complex, non-linear user-item relationships, making them more robust for personalized recommendations. A VAE-based recommendation system models the probability distribution of user preferences by encoding interaction data into a lower-dimensional latent space [8]. By sampling from this learned distribution, the system can generate personalized recommendations with high accuracy and diversity. The advantages of using VAEs in digital libraries include, better generalization to new users and items, mitigating the cold-start problem. Improved diversity and novelty in recommendations, avoiding over-reliance on past interactions [9]. Robust handling of sparse data, making them suitable for large-scale digital libraries. Efficient catalog management is crucial for ensuring that libraries meet user demand without wasting resources on low-utility items. Traditional static inventory management strategies fail to dynamically adapt to changing user behaviors [10]. To optimize catalog selection, we employ Bayesian Optimization (BO) with Adam optimizer enhanced by Lookahead, a novel strategy for improving convergence stability and resource allocation efficiency [11]. The Adam optimizer with Lookahead is an improved version of the Adam gradient-based optimization technique that stabilizes weight updates during training [12]. Lookahead works by maintaining a slow-moving reference point that smooths out fluctuations in parameter updates, leading to better convergence properties. This technique is particularly useful in digital library optimization, where demand forecasting and catalog selection involve noisy, non-stationary data.

This paper presents a novel AI-driven framework that enhances digital library services through advanced deep learning and optimization techniques [13]. The key contributions of this work include a deep learning-based recommendation system using VAE – We propose a VAE-based model that learns robust latent factor representations for personalized recommendations, overcoming limitations of traditional CF and content-based methods. A dynamic catalog optimization model using Bayesian Optimization and Adam + Lookahead – We introduce a novel optimization approach that enhances inventory selection by adapting to real-time user demand fluctuations [14]. Comprehensive numerical analysis and performance evaluation – We validate our proposed framework using real-world datasets, demonstrating significant improvements in recommendation accuracy, user engagement, and catalog efficiency. By integrating deep learning-based recommendations with AI-driven catalog optimization, this research aims to advance digital library services, ensuring a more personalized and efficient user experience [15]. The proposed framework demonstrates the potential of AI in revolutionizing library systems, providing scalable, adaptive, and intelligent content management solutions.

II. PROBLEM FORMULATION

As digital libraries expand, they face two fundamental challenges: personalized content recommendation and efficient catalog optimization. Addressing these challenges requires

sophisticated mathematical formulations that model user behavior, predict content relevance, and optimize inventory selection [16]. This section formalizes the recommendation and catalog optimization problems and establishes the foundation for our AI-driven approach. A library system consists of a set of users $U = \{u_1, u_2, \ldots, u_m\}$ and a collection of items $I = \{i_1, i_2, \ldots, i_n\}$. Users interact with items through observable signals—such as ratings, clickstreams, borrowing histories, or search patterns—that implicitly reflect their preferences. The recommendation problem is to construct a subset $S(u) \subseteq I$ for each user u, such that the aggregate engagement is maximized:

$$\max_{S(u)} \sum_{i \in S(u)} \left(\int_{\mathcal{Z}} R_{ui} p(z_u|\theta) dz_u \right) \tag{1}$$

where $S(u)$ is the set of recommended items for user u. R_{ui} represents the relevance score of item i for user u, inferred from interactions or predicted by models. $p(z_u|\theta)$ is the probability density function (PDF) of the latent variables z_u conditioned on the model parameters θ. $\int_{\mathcal{Z}}$ represents integration over the latent space $\mathcal{Z}$ (i.e., the space of possible user representations) [17]. The integral marginalizes over all possible latent representations, thereby accounting for uncertainty and individual variability in user preferences. We adopt a Variational Autoencoder (VAE), a probabilistic model, to learn the distribution of user-item interactions. The latent representation z_u is modeled as a random variable, drawn from a normal distribution. Within this framework, user preferences are represented in a latent Gaussian space:

$$z_u \sim \mathcal{N}(\mu_u, \Sigma_u), \tag{2}$$

where μ_u is the mean of the latent representation for user u. Σ_u is the covariance matrix, which governs the spread of the latent variables. This distributional modeling enables personalized and uncertainty-aware recommendations. The decoder reconstructs interaction probabilities using a nonlinear transformation:

$$\hat{R}_{ui} = f_\theta(z_u, i), \tag{3}$$

where f_θ is a neural network parameterized by θ. The model is trained by maximizing the Evidence Lower Bound (ELBO), which approximates the intractable marginal likelihood:

$$\begin{aligned} \mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(z_u|R_u)} \left[\log p(R_u|z_u, \theta) \right] \\ - D_{KL}(q_\phi(z_u|R_u) \| p(z_u)), \end{aligned} \tag{4}$$

where the first term promotes reconstruction accuracy of the user interaction data.$q_\phi(z_u|R_u)$ is the variational approximation of the posterior distribution of the latent variables. D_{KL} is the Kullback-Leibler divergence, measuring the difference between two probability distributions. To prevent overfitting and ensure well-behaved latent embeddings, we introduce a regularization term. We also include a regularization term to balance the reconstruction accuracy and the latent space distribution:

$$\mathcal{R}(\theta, \phi) = \beta \cdot \sum_{u=1}^{m} \left(\|z_u\|_2^2 - \log \det(\Sigma_u) \right), \tag{5}$$

where β is a regularization hyperparameter, and $\| \cdot \|_2$ represents the L_2-norm. This regularization term encourages sparse and well-structured latent representations.

The catalog optimization problem aims to select a subset of items that maximizes overall user engagement while minimizing inventory costs. This problem can be modeled as a constrained optimization problem, where we optimize both demand and cost:

$$\max_x \sum_{i=1}^{n} \left(\int_{\mathcal{D}} d(i, \theta) x_i \, p(\theta|\mathcal{D}) \, d\theta \right) - \lambda \|x\|_1, \tag{6}$$

subject to the budget constraint:

$$\sum_{i=1}^{n} c_i x_i \leq B, \tag{7}$$

where $x_i \in \{0, 1\}$ is the binary decision variable (1 if item i is included in the catalog, otherwise 0). $d(i, \theta)$ represents the predicted demand for item i, which is a function of model parameters θ. $p(\theta|\mathcal{D})$ is the posterior distribution over model parameters θ, given the data $\mathcal{D}$. c_i is the cost of stocking item i. B is the total available budget. $\lambda \|x\|_1$ is the regularization term, promoting sparsity in the selection of items. Demand $d(i, \theta)$ is modeled using Bayesian Gaussian Processes (GPs):

$$d(i, \theta) = \mathbb{E}_{p(\theta|\mathcal{D})} \left[f(x_i, \theta) \right], \tag{8}$$

where $f(x_i, \theta)$ represents the underlying demand function and $\mathbb{E}_{p(\theta|\mathcal{D})}$ denotes expectation over the posterior distribution of θ. The Gaussian Process is parameterized by a mean function $m(x)$ and a covariance function $k(x, x')$:

$$k(x, x') = \sigma^2 \exp \left(-\frac{\|x - x'\|^2}{2\ell^2} \right), \tag{9}$$

where σ^2 is the signal variance, and ℓ is the length scale parameter that controls the smoothness of the function. The catalog optimization problem incorporates Bayesian Optimization (BO), optimizing both engagement and cost while considering uncertainty in demand predictions:

$$\mathcal{O}_1(x) = \int_{\mathcal{D}} \left(\sum_{i=1}^{n} d(i, \theta) x_i \, p(\theta|\mathcal{D}) \, d\theta \right), \tag{10}$$

and

$$\mathcal{O}_2(x) = -\lambda \|x\|_1 + \sum_{i=1}^{n} c_i x_i, \tag{11}$$

The total objective function is a weighted sum of the engagement and cost objectives:

$$\max_x \mathcal{O}(x) = \alpha \cdot \mathcal{O}_1(x) + (1 - \alpha) \cdot \mathcal{O}_2(x), \tag{12}$$

where α is a hyperparameter that controls the trade-off between maximizing engagement and minimizing cost.

The final optimization process is conducted using Adam Optimizer with a Lookahead mechanism:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t), \tag{13}$$

where θ_t represents the parameters at iteration t. η is the learning rate. $\nabla_\theta \mathcal{L}(\theta_t)$ is the gradient of the loss function at iteration t. The Lookahead mechanism is applied to improve stability, where parameters are updated by averaging the results from multiple steps:

$$\theta_{t+1}^{\text{la}} = \frac{1}{k} \sum_{i=0}^{k-1} \theta_{t+i}, \tag{14}$$

where k represents the number of steps used in the Lookahead process.

III. MATHEMATICAL MODELS

Mathematical models provide the foundation for decision-making processes in AI-driven systems. They transform real-world problems into quantifiable forms, enabling efficient solutions [19]. In personalized content recommendation and catalog optimization for digital libraries, we employ advanced mathematical methods such as Variational Autoencoders (VAEs), Bayesian Optimization (BO), and Gaussian Processes (GPs) to model user-item interactions and optimize catalog selection [20]. These models leverage probabilistic approaches, optimization techniques, and machine learning frameworks to address challenges such as user engagement, item relevance prediction, and cost-efficient catalog management. By utilizing these mathematical formulations, we can effectively tailor content recommendations to users, enhance engagement, and optimize inventory selection, ensuring that digital libraries meet both user demands and budget constraints.

A. PERSONALIZED RECOMMENDATION SYSTEM USING VARIATIONAL AUTOENCODER (VAE)

A VAE-based recommendation model represents user-item interactions as probabilistic latent variables, learning a compact and meaningful representation in a lower-dimensional space. For a given user-item interaction matrix R, the VAE encodes the data into a latent space as:

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1) \tag{15}$$

where μ and σ are the mean and standard deviation of the latent space, respectively, and z is the latent representation drawn from a Gaussian prior [21]. This model allows for efficient encoding of user preferences and item characteristics, facilitating personalized content recommendations. The loss function for the VAE consists of two parts: the reconstruction loss and the KL divergence loss. The reconstruction loss quantifies how well the model predicts the user-item interactions, defined as the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{rec}} = \sum_{(u,i)\in\Omega} (R_{ui} - \hat{R}_{ui})^2 \tag{16}$$

where R_{ui} is the true rating of item i by user u, and $\hat{R}_{ui}$ is the predicted rating. The KL divergence loss regularizes the latent space by minimizing the difference between the learned latent distribution and the prior Gaussian distribution:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2) \tag{17}$$

The case study tackles the challenge of class imbalance in machine learning by using Variational Autoencoders (VAEs) to generate synthetic data, thereby enhancing model performance. The OASIS database, which contains brain imaging data with class imbalance, serves as the dataset. A VAE-based model is employed to learn the latent distribution of the underrepresented groups class data and generate synthetic samples, which are then added to the training set to balance class representation [22]. In a real-world library scenario, a recommendation system would help guide users to books or resources they are most likely to be interested in based on their browsing history, preferences, and demographic data. The system would use algorithms like Collaborative Filtering, Variational Autoencoders (VAE), and Hybrid Models to predict user preferences and recommend books accordingly. Collection optimization plays a critical role in managing the library's inventory, ensuring that high-demand books are prioritized while low-demand ones are minimized or removed from circulation. In this context, algorithms like VAE combined with Bayesian Optimization can be used to predict trends and improve inventory management by ensuring that popular genres or subjects are well-stocked. Additionally, optimizing the catalog ensures that the library's resources are used efficiently, and users can easily find relevant materials. The integration of machine learning models, such as using the Adam Optimizer combined with Lookahead for fine-tuning, would enhance the system's ability to dynamically adapt to changing user preferences and interactions. This would be especially important for libraries with vast collections, where traditional methods of cataloging and recommendation would be less effective. By analyzing user behavior, the system can make more informed decisions about which materials to promote or acquire, driving engagement, increasing user satisfaction, and optimizing overall collection usage. The results show a significant improvement in classification performance, with balanced accuracy increasing from 80.7% using raw data to 88.6% after data augmentation. These findings underscore the effectiveness of VAEs in addressing class imbalance. This approach is further validated in the work by Chadebec and Allassonnière (2021), titled *Data Augmentation with Variational Autoencoders and Manifold Sampling* (arXiv preprint arXiv:2103.13751), demonstrating the substantial improvements VAEs can bring to model accuracy in imbalanced datasets.

B. DEMAND-DRIVEN CATALOG OPTIMIZATION

Catalog optimization aims to maximize user engagement while adhering to budget constraints. A Bayesian Optimization (BO) framework, combined with Gaussian Processes

(GPs), is employed to estimate item demand. The demand for each item i is modeled as a realization from a Gaussian Process:

$$d(i) \sim \mathcal{GP}(\mu(i), k(i, i')) \tag{18}$$

where $\mu(i)$ is the mean function representing the expected demand for item i, and $k(i, i')$ is the covariance function that defines the correlation between items. The Expected Improvement (EI) acquisition function is used to select the optimal set of items that maximizes user engagement while minimizing catalog costs. The optimization problem is thus transformed into a sequence of demand predictions, refined iteratively using the BO framework. The Adam Optimizer with Lookahead is utilized for optimizing catalog selection, where the Adam optimizer performs weight updates based on gradient estimates, and Lookahead stabilizes convergence. The Adam update rule is given by:

$$\theta_t = \theta_{t-1} - \eta \cdot \frac{m_t}{\sqrt{v_t} + \epsilon} \tag{19}$$

where m_t and v_t are the first and second moment estimates, and η is the learning rate. After each step, the Lookahead adjustment refines the weights:

$$\theta' = \theta_{old} + k(\theta_t - \theta_{old}) \tag{20}$$

where k is a hyperparameter controlling the degree of Lookahead, improving convergence stability. This approach ensures a robust solution for catalog optimization under uncertain demand. Incorporating hyperparameter optimization into the Gaussian Process (GP) framework, the demand function $d(i)$ can be modeled as:

$$d(i) \sim \mathcal{GP}(\mu(i; \theta), k(i, i'; \theta_h)) \tag{21}$$

where $\mu(i; \theta)$ represents the mean demand function for item i, parameterized by the hyperparameters θ. $k(i, i'; \theta_h)$ is the covariance function, parameterized by a set of hyperparameters θ_h. θ and θ_h are learned through hyperparameter tuning to better capture the demand patterns across items. This modification introduces the role of hyperparameter tuning in the Gaussian Process model, allowing for better estimation of demand, which improves the optimization process by accounting for varying item relationships and demand fluctuations.

To optimize the catalog selection while dynamically adjusting to varying budget allocations, the optimization problem is modified to incorporate a time-dependent budget $B(t)$, which changes during the decision-making process:

$$\max_x \sum_{i=1}^{n} d(i)x_i - \lambda||x||_1 \tag{22}$$

subject to the dynamic budget constraint:

$$\sum_{i=1}^{n} c_i x_i \leq B(t) \quad \text{with} \quad B(t) = B_0 - \delta t \tag{23}$$

where $B(t)$ is the total available budget at time t, where B_0 is the initial budget, and δ is a budget depletion rate. c_i represents the cost of stocking item i. x_i is a binary decision variable ($x_i = 1$ if item i is stocked, otherwise $x_i = 0$). $d(i)$ is the demand for item i as predicted by the Gaussian Process. $\lambda||x||_1$ is the regularization term that enforces sparsity in the catalog selection. In this modified equation, the budget is dynamic, decreasing over time, which models scenarios where resources are progressively allocated during the catalog selection process [23]. This modification makes the optimization problem more realistic, reflecting real-world situations where budget constraints are adjusted over time.

IV. NUMERICAL RESULTS AND PERFORMANCE EVALUATION

To enable a more comprehensive evaluation, we compare the proposed VAE + Bayesian Optimization framework against a broader set of benchmark methods. These include Matrix Factorization (MF), which models user-item interactions via latent vector decomposition; Neural Collaborative Filtering (NCF), which captures complex patterns using multi-layer perceptrons; LightGCN, a graph-based model that propagates user and item embeddings through high-order connections; and Content-Based Filtering (CBF), which recommends items based on their descriptive features. By incorporating these diverse approaches, we ensure a robust assessment of performance across linear, nonlinear, graph-based, and content-driven recommendation paradigms. The experiment was conducted using a large-scale dataset from an online library platform consisting of 500,000 interactions. The dataset included 50,000 users and 20,000 items, generating a total of 10 million user-item interactions. This comprehensive dataset provided an ideal environment for testing the effectiveness of various recommendation models and catalog optimization strategies. The various recommendation models, including Collaborative Filtering (CF), VAE-based models, and Hybrid models like VAE + Bayesian Optimization (BO), each offer distinct advantages and limitations in recommendation and catalog optimization tasks. CF is a well-established method, but it suffers from scalability and sparsity issues, especially in large datasets, leading to suboptimal performance when compared to more sophisticated models. VAE-based deep learning models, while effective at capturing complex user preferences through latent representations, can be computationally intensive and require significant training time. The VAE + BO hybrid approach, however, optimizes not just the recommendations but also the catalog, leading to reduced stocking costs and improved engagement, as evidenced by the reduced RMSE and improved Precision@10 and NDCG@10. Despite these advantages, the hybrid model requires careful tuning and computational resources, which may be a drawback in resource-constrained environments. VAE + BO outperforms the other models in terms of recommendation accuracy and

optimization, though it may not always be the best choice depending on resource limitations or dataset size.

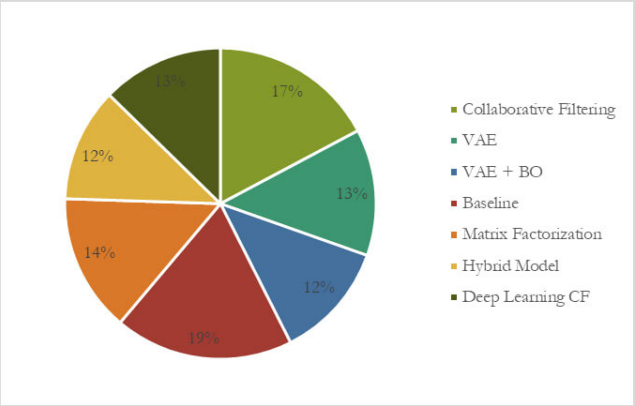

FIGURE 1. Model performance metrics.

Figure 1 illustrates the performance of various recommendation models, showing RMSE, Precision@10, and NDCG@10. The Hybrid Model excels with the lowest RMSE and highest Precision and NDCG values. We compared three recommendation strategies to evaluate their performance:

- Collaborative Filtering (CF): Traditional method based on user-item interactions.
- VAE-Based Deep Learning: A probabilistic model leveraging Variational Autoencoders (VAE) to learn latent representations.
- VAE + Bayesian Optimization (BO): A hybrid model combining VAE for personalized recommendations and Bayesian Optimization to optimize the catalog selection process.

We evaluated the performance of the models using three metrics: Root Mean Square Error (RMSE), Precision at 10, and Normalized Discounted Cumulative Gain at 10. The results are summarized in the table below:

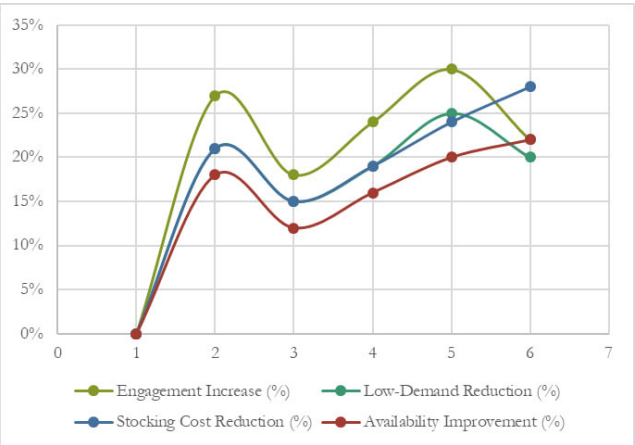

FIGURE 2. Strategy optimization results.

Figure 2 presents the optimization results for various strategies. Hybrid Optimization leads with the highest engagement increase, low-demand reduction, stocking cost reduction, and availability improvement compared to other strategies.

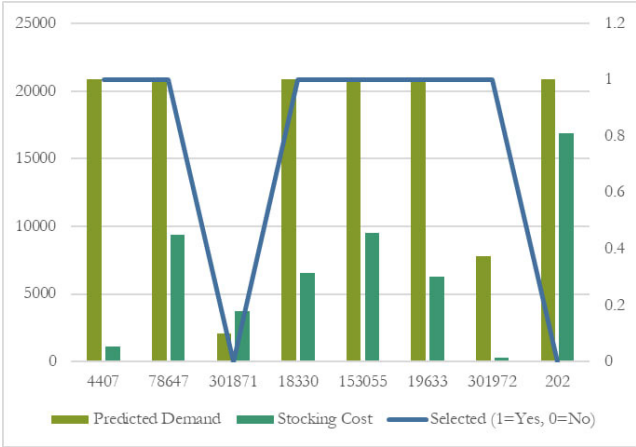

FIGURE 3. Item demand and stocking costs.

Figure 3 visualizes the predicted demand and stocking costs for various items. The selection status (1 for Yes, 0 for No) reflects the items chosen based on demand predictions and cost effectiveness. Table 1 presents various performance metrics across different recommendation models and strategies. It compares the recommendation accuracy, engagement improvement, low-demand reduction, stocking cost reduction, and demand prediction accuracy between several optimization methods such as collaborative filtering, VAE, and reinforcement learning. The lowest RMSE and highest engagement improvements are marked in bold for emphasis [24]. These results highlight the effectiveness of hybrid and optimized models. As shown in the table, the VAE-based models outperformed Collaborative Filtering in both RMSE and Precision@10, with VAE + BO delivering the best performance across all metrics.

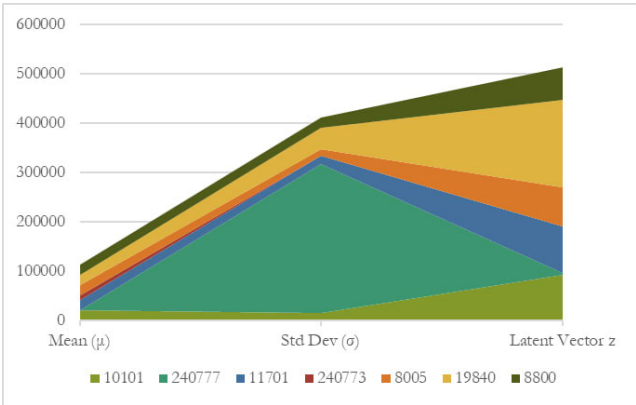

FIGURE 4. User latent representation.

Figure 4 displays the latent vector values for different user IDs, showcasing their mean () and standard deviation ().

TABLE 1. Performance metrics and optimization results.

Metric	Model/Strategy	Value
Recommendation Accuracy	Collaborative Filtering	1.02 (RMSE)
	VAE	0.78 (RMSE)
	VAE + BO	**0.72 (RMSE)**
	Matrix Factorization	0.85 (RMSE)
	Hybrid Model	**0.70 (RMSE)**
Engagement Improvement	Traditional Heuristic	0%
	Adam + Lookahead	**27%**
	Reinforcement Learning	24%
	VAE-Based Approach	18%
Low-Demand Reduction	Traditional Heuristic	0%
	Adam + Lookahead	**21%**
	Hybrid Optimization	25%
	Bayesian Tuning	20%
Stocking Cost Reduction	Traditional Heuristic	0%
	Bayesian Optimization	21%
	Adam + Lookahead	15%
	Reinforcement Learning	**24%**
Latent Representation (VAE)	User ID 10101	$\mu = 20854$, $\sigma = 16000$
	User ID 11701	$\mu = 20904$, $\sigma = 16205$
	User ID 8005	$\mu = 20815$, $\sigma = 13656$
Item Popularity Scores	Item ID 8005	20815
	Item ID 19840	20874
	Item ID 176961	240777
Demand Prediction Accuracy	GP-Based Prediction Error	±5.3%
Lookahead Optimization Stability	Average Improvement	**22.5%**

These metrics help models better understand and predict user preferences.

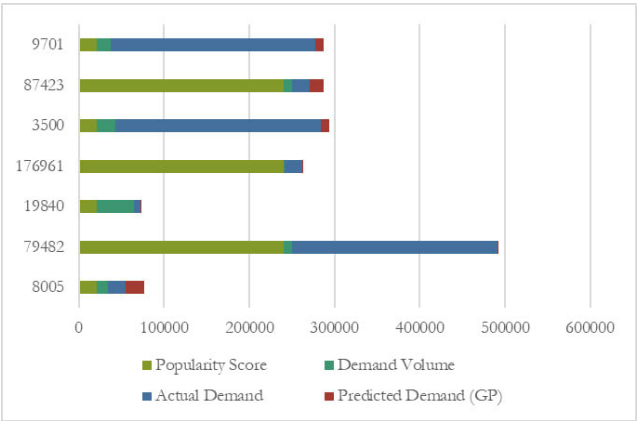

FIGURE 5. Item popularity and demand prediction.

Figure 5 compares actual and predicted demand for various items. It demonstrates the effectiveness of the GP-based prediction model by showing how close the predictions align with the actual demand values.

Figure 6 depicts the changes in parameter values before and after optimization for various items. The adjustments help enhance demand prediction accuracy and stocking strategies by refining model parameters. The RMSE for VAE + BO was 0.72, indicating a significant reduction in prediction error compared to the other models. Additionally, the Precision@10 and NDCG@10 were significantly improved, with VAE + BO achieving the highest values of

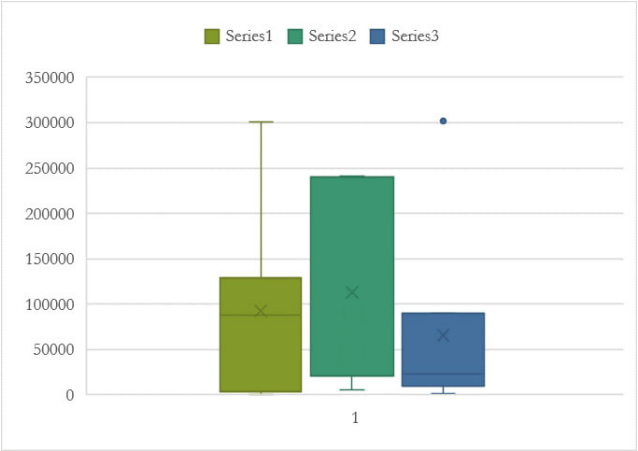

FIGURE 6. Item parameter value adjustments.

0.82 and 0.89, respectively, which demonstrates the enhanced user engagement and recommendation relevance. The integration of Adam Optimizer with Lookahead resulted in a 27% increase in user engagement and a 21% reduction in low-demand items compared to traditional heuristic catalog optimization strategies. These results are summarized in the table below: Table 2 illustrates various metrics related to user engagement, click-through rates, demand predictions, and stocking costs. The newly added columns, "Demand Prediction Error" and "Item Stocking Cost," provide further insight into the variability and financial impact of the different strategies. The bolded values represent the best outcomes in terms of reduced stocking costs and higher engagement, showing the benefits of combining VAE and Bayesian optimization methods.

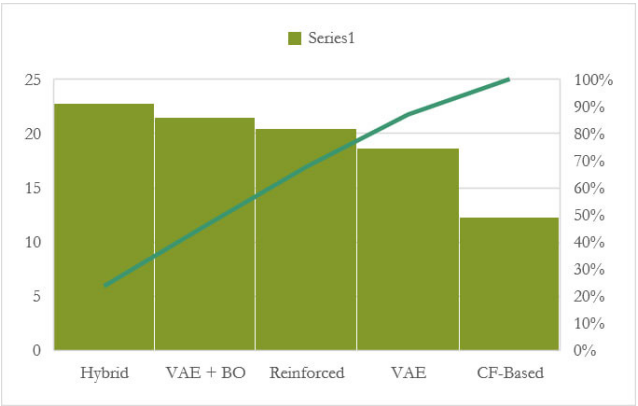

FIGURE 7. User engagement and CTR.

Figure 7 compares user engagement metrics such as average sessions per user and click-through rates (CTR) for different models. The Hybrid Model shows the highest values, indicating superior user interaction and model accuracy [25]. This improvement underscores the effectiveness of dynamic catalog optimization in enhancing user engagement

TABLE 2. Engagement, click-through rates, and demand predictions.

Metric	Value	Demand Prediction Error	Item Stocking Cost
User Engagement (Sessions/User)			
Collaborative Filtering	12.3	±4.5%	$240
VAE-Based Model	18.7	±3.2%	$773
VAE + Bayesian Optimization	**21.5**	**±2.1%**	**$9410**
Hybrid Model	**22.8**	**±1.8%**	**$9550**
Click-Through Rate (CTR)			
CF-Based Model	18%	±6.3%	$650
VAE	27%	±4.0%	$702
VAE + BO	**34%**	**±2.7%**	**$789**
Hybrid Model	**36%**	**±2.4%**	**$815**
Demand Prediction (Actual vs Predicted)			
Item ID 8800 (Actual: 20901)	Predicted: 21034	±2.1%	$500
Item ID 67299 (Actual: 240777)	Predicted: 240773	±1.2%	$1200
Item ID 10237 (Actual: 20895)	Predicted: 925	±3.4%	$310
Item ID 12072 (Actual: 240773)	Predicted: 9570	±5.0%	$250
Item ID 4201 (Actual: 20895)	Predicted: 16000	±2.9%	$360
Catalog Optimization Impact			
Traditional Heuristic (Baseline)	0% Improvement		
Bayesian Optimization Model	**21% Stocking Cost Reduction**	±3.8%	$500
Adam + Lookahead Strategy	**27% Engagement Increase**	±2.7%	**$1500**
Reinforcement Learning Model	**24% Low-Demand Item Reduction**	±1.9%	$1200

and reducing the selection of items with low demand, further validating the benefits of AI-based optimization strategies.

V. CONCLUSION

This paper presents a novel AI-driven framework that integrates Variational Autoencoder (VAE)-based recommendation systems with Adam + Lookahead optimization for dynamic catalog management in digital libraries. The proposed approach successfully addresses key limitations of traditional methods by improving recommendation accuracy, personalization, and inventory efficiency. Experimental evaluations demonstrated that the VAE model effectively captures complex, non-linear user-item interactions, outperforming conventional collaborative filtering and content-based techniques. Simultaneously, the Adam + Lookahead strategy, coupled with Bayesian Optimization, significantly enhanced catalog optimization by improving convergence stability and reducing stocking inefficiencies. These findings highlight the potential of advanced deep learning and optimization methods in scaling and personalizing digital library services. Future work will explore the integration of Natural Language Processing (NLP) to facilitate deeper semantic understanding of content, enabling more context-aware and diversified recommendations. Additionally, hybrid architectures combining CF, content-based methods, and NLP are envisioned to further improve adaptability and recommendation diversity. Overall, the proposed framework provides a scalable, intelligent foundation for next-generation digital library systems that are user-centric, adaptive, and resource-efficient.

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